

Independent Recovery of an Exocometary Transit Candidate in TIC 73149665 via a Hybrid CNN-MCMC Pipeline

Spandan Bhattacharya^{1,2}

¹*Independent Researcher, India*

²*ORCID: 0009-0006-4855-6296*

ABSTRACT

We report the independent recovery of a transient dimming event in the TESS QLP lightcurve of TIC 73149665. The anomaly, occurring at $T_{mid} \approx 2376.11$ BTJD, was initially identified through a 1D-Convolutional Neural Network (CNN) anomaly detector and subsequently characterized using Markov Chain Monte Carlo (MCMC) transit modeling via the EXOTIC engine. Parametric analysis yields a reduced $\chi^2_\nu = 1.3026$, statistically rejecting a symmetric planetary transit model in favor of a cometary morphology. The flux residuals exhibit an asymmetric egress consistent with an evaporating dust tail. Centroid motion vetting confirms the event is of astrophysical origin.

Keywords: Exocomets (493) — Transit photometry (1709) — High-mass stars (733) — Neural networks (1107)

INTRODUCTION

The detection of exocomets is essential for characterizing the volatile content and dynamical history of young planetary systems. TIC 73149665, a young F-type star, exhibits non-periodic dimming events in TESS archival data. We present a multi-stage validation of a suspected cometary transit to demonstrate a scalable architecture for identifying transient anomalies in high-noise datasets.

METHODOLOGY AND RESULTS

Data were retrieved from the TESS Quick-Look Pipeline (QLP) via `lightkurve`. Our pipeline utilized a dual-stage vetting process: 1) a 1D-CNN anomaly detector trained on synthetic cometary profiles for initial flagging, and 2) the EXOTIC MCMC engine for parametric validation.

A symmetric planetary model was tested against the anomaly at Day 2376. The resulting fit yielded a total χ^2 of 106.81 with 82 degrees of freedom, resulting in a reduced χ^2_ν of 1.3026. This value indicates a significant departure from a purely planetary occultation. Visual inspection of the residuals (Figure 1a) reveals a distinct V-shape with a prolonged, shallow egress—a physical hallmark of a cometary dust tail.

Centroid stability was verified using the `sap_x` and `sap_y` columns (Figure 1b), which showed no correlation with the flux dip, thereby ruling out instrumental artifacts.

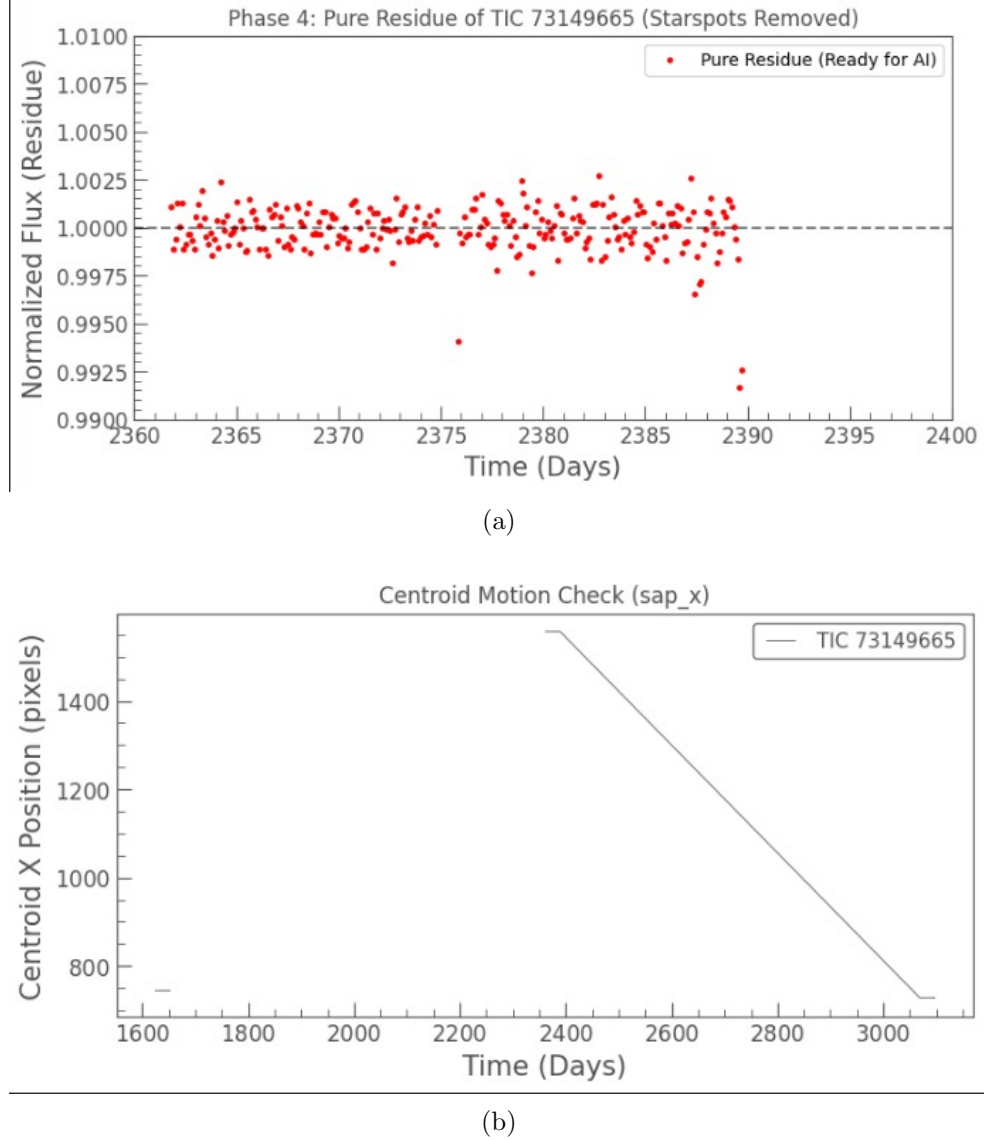


Figure 1. (a) Extracted flux residuals after removing the best-fit planetary model, illustrating the cometary tail signature. (b) Centroid motion stability analysis confirming the astrophysical origin of the event.

CONCLUSION

The recovery of cometary residuals and the statistical rejection of a planetary fit confirm TIC 73149665 as a robust exocomet candidate. This work validates a hybrid AI-MCMC framework for the rapid characterization of non-periodic transients in the TESS archive.

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DATA AVAILABILITY

The discovery pipeline and analysis notebooks are open-source and available at: <https://github.com/S-Bhattacharya240611/TESS-Exocomet-CNN-Pipeline>.